

MODULE 7

Threats to the Computer

Security, Vulnerabilities, and Malware

| DEFINITION: MALWARE

Malicious Software (Malware): Any software specifically designed to disrupt, damage, or gain unauthorized access to a computer system.

Key Objectives

- ✓ Financial Theft
- ✓ Surveillance
- ✓ Denial of Service

| THE CLASSIC VIRUS



| COMPUTER WORMS

- ✓ **Self-Replicating:** Does NOT require a host file to spread.
- ✓ **Network-Based:** Travels through internet connections and local networks.
- ✓ **Impact:** Consumes massive bandwidth and creates backdoors for hackers.

| TROJAN HORSES



Disguise

Appears to be a useful tool (calculator, game) but contains hidden malicious code.



Backdoors

Allows an attacker remote control over the victim's machine.

| RANSOMWARE

\$1B+

ESTIMATED ANNUAL LOSS

Encrypts user data and demands payment for the decryption key.

| SPYWARE & ADWARE

Spyware

Secretly monitors browsing habits and captures personal data without consent.

Adware

Floods the system with unwanted pop-up advertisements, often slowing performance.

| KEYLOGGERS

Keystroke Recording

Capturing every single key pressed on the keyboard.

Identity Theft

Used specifically to harvest passwords and credit card numbers during typing.

| LOGIC BOMBS

A string of malicious code that lies dormant until a specific condition is met, such as a specific date or a user being deleted from a database.

Trigger Examples

- ✓ Friday the 13th
- ✓ System Admin dismissal

| ROOTKITS

Invisibility

Modifies the operating system to hide its presence from Antivirus software.

Admin Level

Grants the attacker full "root" or administrative privileges over the machine.

| HOW THREATS SPREAD



| SIGNS OF INFECTION

Symptom	Probable Cause
System Slowdown	Malware background processes
Blue Screen (BSOD)	Rootkit / Logic Bomb damage
Missing Files	Worm / Virus deletion
Unknown Toolbars	Adware installation

| PHISHING

Social Engineering: Tricking users into revealing sensitive info by masquerading as a trustworthy entity in electronic communications.

Red Flags

- ✓ Sense of urgency
- ✓ Poor spelling/grammar
- ✓ Mismatched URLs

| DOS AND DDOS ATTACKS

HTTP Flood

High Volume

Botnet Size

1M+ Zombies

Overwhelming a server's resources to make a website or service unavailable to legitimate users.

| MAN-IN-THE-MIDDLE (MITM)

The attacker secretly relays and possibly alters the communications between two parties who believe they are directly communicating with each other.

Common Venue

Unsecured Public Wi-Fi hotspots at airports or cafes.

| ANTIVIRUS SOFTWARE

Signatures

Identifying malware by comparing files against a database of known threats.

Heuristics

Detecting threats by observing suspicious behavior (behavioral analysis).

| FIREWALLS

- ✓ **The First Line:** Monitoring and controlling incoming/outgoing network traffic.
- ✓ **Packet Filtering:** Inspecting every packet of data for security rules.
- ✓ **Hardware vs Software:** Protecting both the entire network and individual devices.

| ENCRYPTION

HTTPS

SECURE BROWSING

Using SSL/TLS protocols to ensure data cannot be read if intercepted.

| THE HUMAN ELEMENT



Strong Passwords

Using unique, long, and complex phrases for every account.



2FA / MFA

Adding a second layer of verification (SMS, Authenticator App).